Data Cleaning, Transformation and Modelling

Question 1 : What is Data Loading in Power BI and why is it considered the first step of analysis?

- Data Loading in Power BI is the process of importing data from different sources, such as Excel files, SQL databases, CSV files, and web sources, into Power BI for analysis and visualization.

Why is it considered the first step of analysis?

1. Provides data: It brings the required data into Power BI for analysis.
2. Enables data cleaning: After loading or connecting to data, we can use Power Query to remove duplicates, handle missing values, and correct errors.
3. Supports data modelling: The loaded data can be connected through relationships between tables.
4. Enables visualization: It allows users to create charts, tables, dashboards, and reports.
5. Helps in decision-making: It makes data available for identifying trends and generating business insights.

Example: A sales manager imports an Excel file containing product sales, prices, and regions into Power BI to analyze regional sales performance.

Question 2 : Explain the difference between “Load” and “Transform Data” in Power BI.

- Difference between “Load” and “Transform Data” in Power BI

1. Load:

- The Load option imports data directly into Power BI without making changes in Power Query.
- The data becomes available for creating reports, charts, and dashboards.
- Example: Importing an Excel sales file directly into Power BI.

2. Transform Data:

- The Transform Data option opens Power Query Editor, where we can clean and modify data before using it.
- We can remove duplicates, handle missing values, change data types, rename columns, and filter rows.
- Example: Removing duplicate sales records and correcting date formats before loading the data.

Load	**Transform Data**

Imports data directly into Power BI.	Opens Power Query Editor.
Does not require additional transformation steps.	Allows cleaning and modifying data.
Best for data that is already prepared.	Best for data that needs preparation.

Question 3 : What is a Fact Table and a Dimension Table? Give examples from the dataset.

- 1. Fact Table:
- A Fact Table stores measurable numerical data about business activities, such as sales amount, quantity sold, profit, and revenue. It contains foreign keys that connect it to dimension tables.
- Example: In a sales dataset, a fact table may contain ProductID, RegionID, Quantity Sold, and Sales Amount.
- 2. Dimension Table:
- A Dimension Table stores descriptive information that helps us understand and analyze the facts, such as product names, customer details, dates, and regions.
- Example: A product dimension table may contain ProductID, ProductName, Category, and Price.

Fact Table	**Dimension Table**
Stores numerical measurements.	Stores descriptive information.
Contains foreign keys.	Usually contains primary keys.
Example: Sales Amount, Quantity Sold, Profit.	Example: Product Name, Region, Date.
Used to calculate business performance.	Used to filter, group, and analyze data.

Question 4 : Why is Star Schema preferred over Snowflake Schema in Power BI?

- A Star Schema is a data model in which one central fact table is connected directly to multiple dimension tables. It is generally preferred over a Snowflake Schema in Power BI because it is simpler to understand and often provides better query performance.

Reasons why Star Schema is preferred:

1. Better performance: Fewer table joins can improve query performance.
2. Simple data modelling: Relationships between fact and dimension tables are easier to create and manage.
3. Easy report creation: Users can easily analyze data using fields from dimension tables.
4. Simpler DAX calculations: Measures and calculations are generally easier to write and maintain.
5. Easy to understand: Its structure is straightforward for developers and business users.

Example:

- FactSales: SalesID, ProductID, RegionID, SalesAmount
- DimProduct: ProductID, ProductName, Category
- DimRegion: RegionID, RegionName
- DimDate: DateID, Date, Month, Year

In a Star Schema, all three dimension tables connect directly to the central FactSales table.

Star Schema	**Snowflake Schema**
Dimension tables are generally denormalized.	Dimension tables are split into related sub-tables.
Fewer joins are usually required.	More joins may be required.
Simpler to understand and maintain.	More complex to understand and maintain.
Generally preferred for Power BI reporting.	Useful when normalized dimensions are needed.